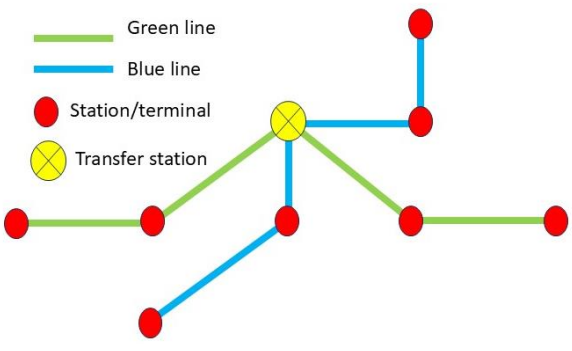
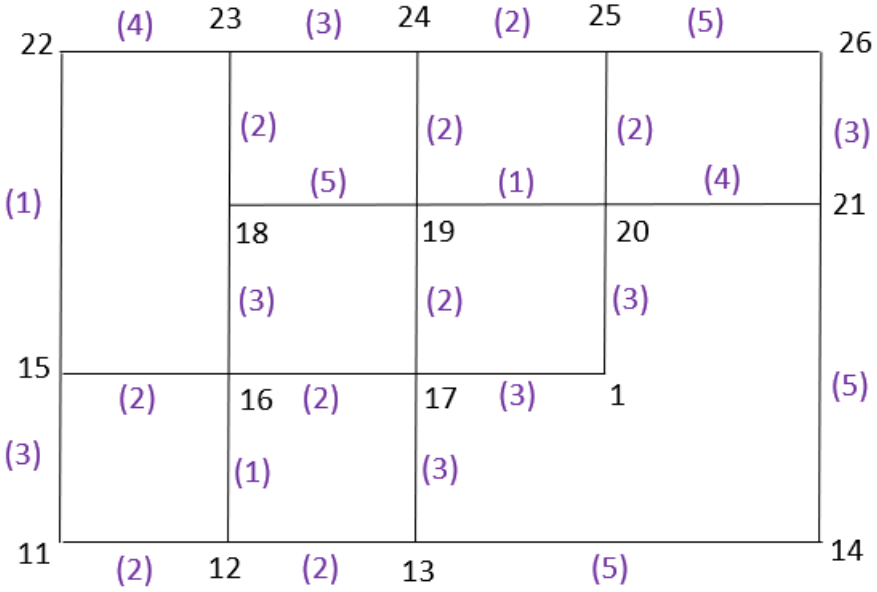
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

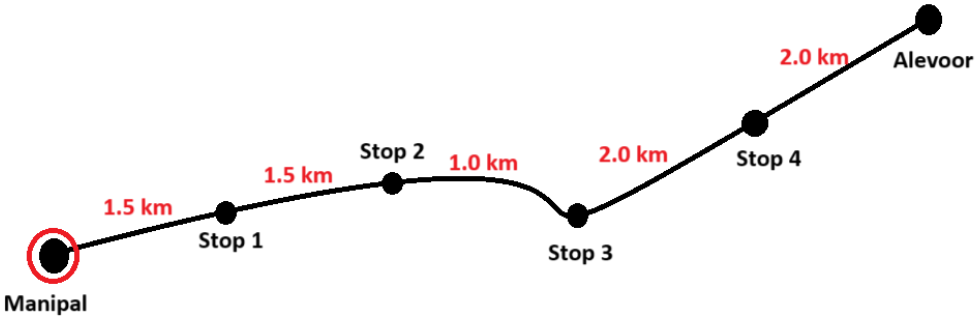
DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION - REGULAR		**November, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**PE - III**
Course Name	**Urban Mass Transportation System**	**Course Code**	**CIE 4067**
Semester	**VII**	**Max Marks**	**50**
Date	**29/11/2024**	**Duration**	**3 hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	A transit authority must design a rapid rail line to meet peak-hour demand of 10000 passengers/hr, with a required speed of 10.67 to 12.19 m/sec. Following are the assumptions made: Deceleration 0.60 m/sec^2, safety factor k =1.35, min headway= 120sec, max headway = 240sec, load factor = 0.9, guideway utilization factor=0.6, station platform limit =10 vehicles, car length=21.34m, car capacity=130 passengers. How many cars should a train consist, to provide adequate passenger volume capacity? What will be the corresponding headway?	**5**	2	1, 2, 3, 12	**3**
1B	What are the requirements of a bus loading zone? Explain.	**3**	5	1, 2, 3, 12	**2**
1C	Explain the classification of mass transport systems according to their technology.	**2**	2	1, 2, 3, 12	**2**
2A	Find the network size and topology indicators and write your inference. All the stations are spaced at a distance of 2 km. 	**5**	3	1, 2, 3, 12	**4**

2B	Explain the features of express bus, local bus and light rail transit systems.	**3**	5	1, 2, 3, 12	**3**
2C	Identify the transit line type given in the diagram below and explain its characteristics. 	**2**	3	1, 2, 3, 12	**3**
3A	Determine the shortest path from station 1 to stations 11, 14, 22 and 26 of a bus transit network as shown. The numbers in parenthesis () indicate distance in km. 	**5**	4	1, 2, 12	**4**
3B	Define transit scheduling. Explain its components.	**3**	4	1, 2, 12	**2**
3C	Explain the characteristics of low-frequency transit service and regular-frequency transit service.	**2**	4	1, 2, 12	**2**
4A	Develop a master schedule for the bus transit service running from Alevoor to Manipal using the data below. Average speed of the bus = 30 kmph Headway = 20 minutes Peak service hours: 8:00 am to 9:30 am Other routes converge on Manipal terminal at: 02 and :32 past the hour.	**5**	4	1, 2, 12	**4**

					
4B	Compare between zonal fare and sectional fare systems adopted in transit service.	**3**	5	1, 2, 3, 12	**4**
4C	Why blocking is important in scheduling work? Explain.	**2**	4	1, 2, 12	**2**
5A	Illustrate the levels of service adopted in the planning of the queuing area at bus stops.	**5**	5	1, 2, 3, 12	**3**
5B	Explain the self-service transit fare collection system.	**3**	5	1, 2, 3, 12	**2**
5C	C	**2**	5	1, 2, 3, 12	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***0***
2	***7***
3	***7***
4	***17***
5	***19***

BL	Total Marks
2	**17**
3	**15**
4	**18**
5	